



A human forceful precision grip grasping a bone at the base of the thumb from *Australopithecus africanus*. TRACY KIVELL & MATTHEW SKINNER

How human ancestors got a grip

By [Ann Gibbons](#) | Jan. 22, 2015 , 2:00 PM

Squeeze a baseball or pen between your thumb and the tips of your fingers: You are using what researchers call a precision grip, a highly evolved adaptation thought to be unique to modern humans and our most recent ancestors. Chimpanzees, for example, have thumbs too short to allow them to grip objects so precisely. But a new study suggests that human ancestors in South Africa had a good grip perhaps as early as 3 million years ago—and so may have wielded stone tools earlier than expected.

Human ancestors have been handy since at least 2.6 million years ago, when the earliest reliably identified stone tools appear in the fossil record at Gona, Ethiopia. But the identity of the first toolmaker has been mysterious, because more than one type of hominin was alive 2 million to 3 million years ago: our genus *Homo*, plus several species of *Australopithecus*, including *A. africanus* from South Africa and *A. afarensis*—the species of the famous skeleton of Lucy, who lived near Gona. Stone tools have been found at sites with *Australopithecus* fossils, as well as bones with possible cut marks dating back to 3.2 million to 3.4 million years ago, but in the absence of a fossil hand gripping a tool, it has been impossible to prove that australopithecines made and used tools.

To complicate matters further, complete hand fossils—full of tiny fragile bones—are extremely rare. Instead, paleoanthropologists have had to rely on clues from traits in isolated finger bones, such as a report in 2013 [that our direct ancestor *Homo erectus* used a precision grip 1.7 million years ago](#). And in 1997, [a landmark study of a](#)

Science's extensive COVID-19 coverage is free to all readers. To support our nonprofit science journalism, please [make a tax-deductible gift today](#).

[Get a tip](#)
How to contact the news team

Related Jobs

Assistant Professor

Midwestern University - Downers Grove, Illinois

Assistant Professor

Midwestern University - Downers Grove, Illinois

Tenure Track Faculty

Duke University - Durham, North Carolina

[MORE JOBS ►](#)

ScienceInsider

HHMI fires prominent biologist for sexual harassment

By Meredith Wadman | Aug. 21, 2021

[Do Delta](#)

composite hand skeleton of Lucy's species, *A. afarensis*, suggested that Lucy lacked a full precision grip 3.1 million years ago, but that her species had developed several but not all of the traits in its hand bones that are associated with the precision grip required for habitual toolmaking. This hinted that the species was beginning to use its hands more like us, perhaps for tool use.

A new method for analyzing CT scans of fossils, however, is giving paleoanthropologists a new, more detailed look at the internal structure of hand bones, revealing how their habitual handiwork shaped their hands during life. In the report published online today in *Science*, husband-and-wife team Matthew Skinner and Tracy Kivell, both paleoanthropologists at the University of Kent and the Max Planck Institute for Evolutionary Anthropology, and their colleagues **used CT scanners to examine the pattern of fossilized spongy bone tissue within finger and hand bones of hominins that lived 2 million to 3 million years ago**.

When modern humans use a forceful precision grip frequently during childhood, their bones adapt: Tiny spicules, or filaments, of bony tissue called trabeculae form and act as struts to provide more bone density—and strength—where the forces are greatest. Gripping a tool distributes forces asymmetrically along the base of the thumb and the side of the knucklebones, and modern hands show a corresponding pattern of trabecular density. By contrast, when chimps grip a branch, the forces align across the thumb more evenly, leaving a symmetrical pattern in the trabecular bone.

When the team scanned hand bones from four members of *A. africanus* that lived in South Africa between 2 million and 3 million years ago, they found that the pattern of the trabecular bone was asymmetrical, as in modern humans and Neandertals that use tools frequently (as they also show in their study). This suggests that *A. africanus* was using a “human-like” precision grip “much earlier and more frequently than previously considered,” the authors write. They stop short of saying that *A. africanus* was using and making stone tools and acknowledge that these grips could have been used for a number of different activities with tools. However, the similarity of the pattern with modern humans suggests that hominins had the capacity for stone tool use more than half a million years before such tools are securely dated in the fossil record. They also have scanned hand bones of other members of *Australopithecus*, including Lucy's species, *A. afarensis*, but the pattern of use was not preserved in that species's trabeculae.

The new method provides “solid evidence that *A. africanus* was likely using forceful precision pinch grips,” says ancient hand expert Matthew Tocheri of Lakehead University in Thunder Bay, Canada, and the Smithsonian Institution's Human Origins Program in Washington, D.C. But he notes that the evidence for tool use is still indirect. To show that, researchers will need to find stone tools at many early *Australopithecus* sites. Other activities that the hominins engaged in frequently during development, such as digging tubers or climbing, might also explain the signs of stress, warns paleoanthropologist C. Owen Lovejoy of Kent State University in Ohio.

Future studies will be needed to see if this pattern is found just in *A. africanus* or in other members of *Australopithecus* as well, says paleoanthropologist Brian Richmond of the American Museum of Natural History in New York City. Regardless of the precise cause,

'breakthroughs' really mean vaccine protection is waning, and are boosters the answer?

By Gretchen

Vogel | Aug. 20, 2021



'I don't want to die.' Afghan researchers fear for their safety—and the future of science

By Richard

Stone | Aug. 20, 2021



Scientists map urban heat islands—and track how communities are affected

By Anil

Oza | Aug. 17, 2021



A grim warning from Israel: Vaccination blunts, but does not defeat Delta

By Meredith

Wadman | Aug. 16, 2021



More ScienceInsider

Sifter

That sea snake charging toward you is probably just looking for love

By Rachel

Fritts | Aug. 20, 2021



Saturn has a surprisingly large—and sloshy—core

By Rachel

Fritts | Aug. 18, 2021



Santorini volcano erupts less often when sea levels are high

By Alex

Viveros | Aug. 4, 2021



Soccer position could influence risk of brain disease



Posted in: [Archaeology](#), [Evolution](#), [Paleontology](#)
doi:10.1126/science.aaa6381



Ann Gibbons
Ann is a contributing correspondent for *Science*.
[Twitter](#)

Wild deer have coronavirus antibodies



By Rachel Fritts | Aug. 3, 2021
More Sifter

More from News

[Archaeologists unearth Nazi-era massacre in Poland's 'Death Valley'](#)



[DNA from dirt can offer new view of ancient life](#)



[These ancient weights helped create Europe's first free market more than 3000 years ago](#)



Platform Migration Update
We're moving to a new website.
[Learn more about what is changing and how this will impact you.](#)